The Superconductor-Insulator Transition, Quantum Phase Transitions and Copper Oxides

Based on *Superconductor-insulator transition in* $La_{2-x}Sr_xCuO_4$ *at the pair quantum resistance*, by Bollinger, et al.

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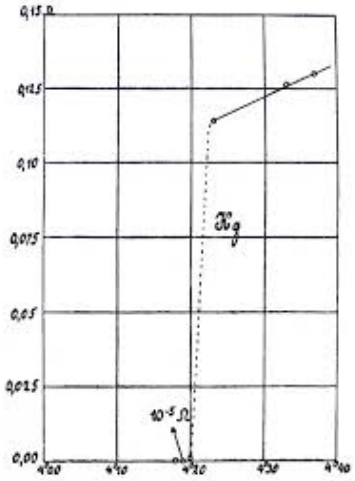
PHY 598, Sept. 15, 2016

Outline

1. Superconductivity
   - Brief History
   - Aspects of Superconductivity
   - High-Temperature and Thin Film Superconductivity
2. Quantum Phase Transitions and the Superconductor - Insulator Transition
   - Classical vs. Quantum Phase Transitions
   - Quantum Statistical Mechanics and Finite-"Size" Scaling
   - Superconductor-Insulator Transition as a QPT
3. Superconductivity in Copper Oxides
   - Fabrication – The Electric Field Effect
   - Scaling Behavior and Collapse

A Brief History of Superconductivity

Discovered in 1911 by Heike Kamerlingh Onnes



Bardeen-Cooper-Schrieffer (BCS) theory developed in 1957

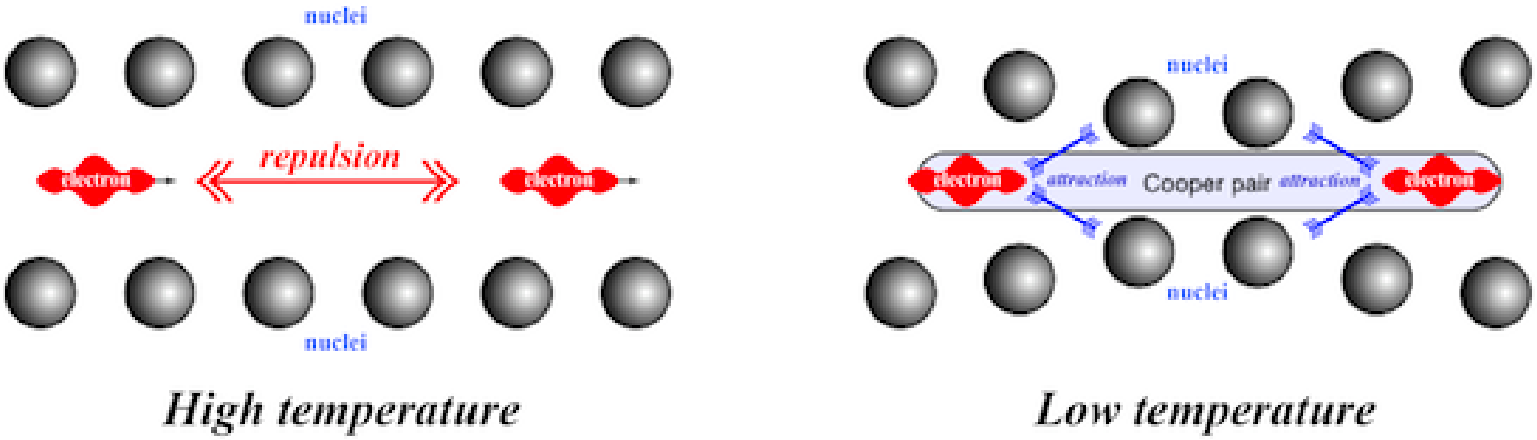
credit: http://archives.library.illinois.edu/archon/?p=digitallibrary/digitalcontent&id=1512

Aspects of Superconductivity

BCS: electron-phonon interaction creates attraction between electrons – allows Cooper pairing

Phase transition described by Landau-Ginsburg theory with a complex order parameter $\Psi = \Delta e^{i\phi}$ (roughly a macroscopic wave-function)

Vanishing resistance requires long-range phase coherence



courtesy Vatche Sahakian

High-Temperature and Thin Film Superconductivity

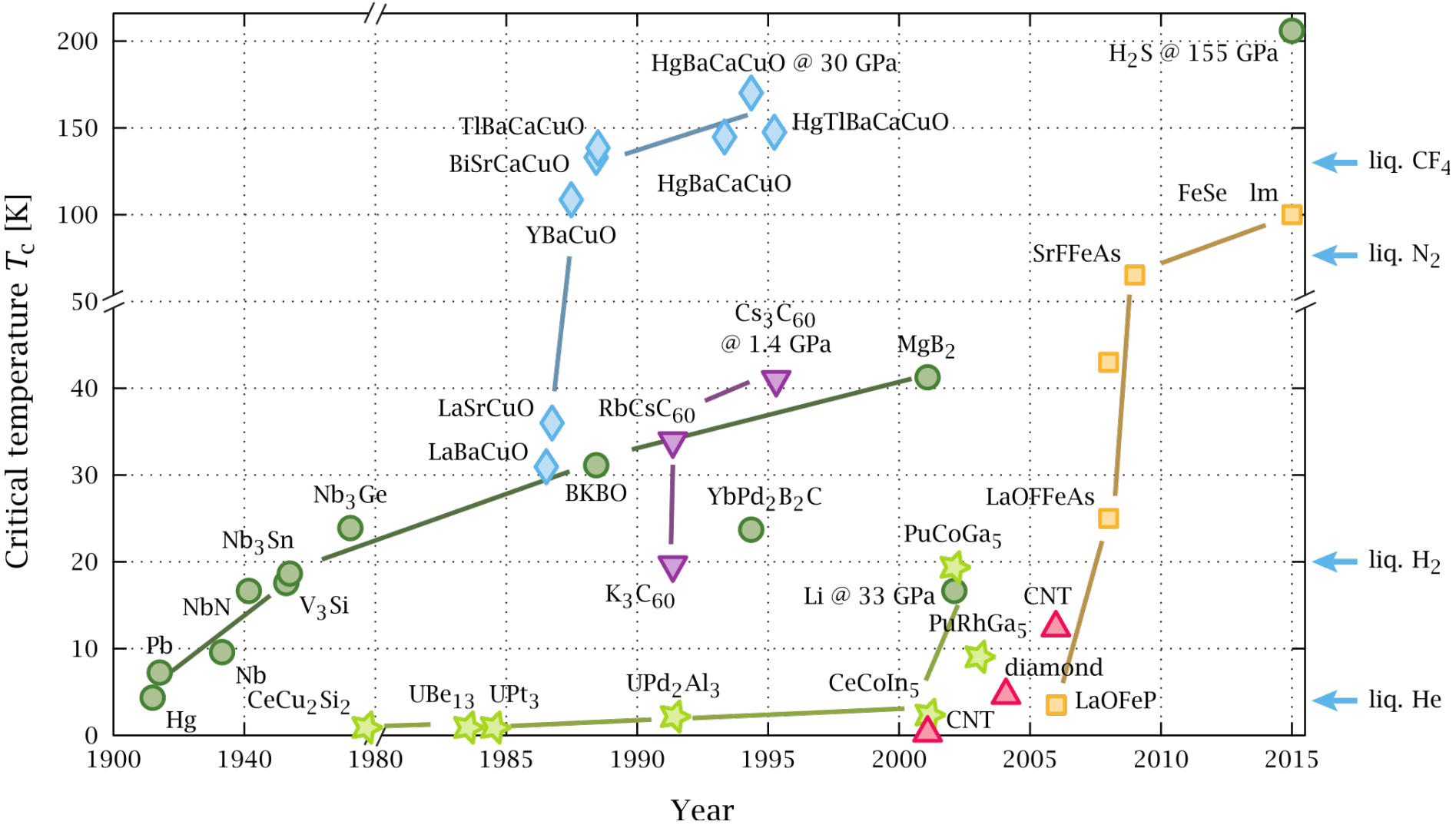
BCS theory implies that T_c can't be higher than ~ 30 K

Furthermore, weakly interacting electron systems are always localized in two dimensions.

Georg Bednorz and K. Alex Müller discovered first High-T_c superconductor in 1986 (LaBaCuO)

Around the same time, theoretical developments demonstrated how superconductivity could still occur in thin films (KBT transitions)

Since the 90's, evidence has been mounting that thin film superconductivity results from a different type of phase transition.


Critical temperature T_c [K]
Year
200
150
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1940
1980
1985
1990
1995
2000
2005
2010
2015
HgBaCaCuO @ 30 GPa
H_2S @ 155 GPa
TlBaCaCuO
HgTlBaCaCuO
BiSrCaCuO
HgBaCaCuO
YBaCuO
FeSe
lm
SrFFeAs
Cs_3C_{60}
@ 1.4 GPa
MgB_2
LaSrCuO
$RbCsC_{60}$
LaBaCuO
BKBO
Nb_3Ge
$YbPd_2B_2C$
LaOFFeAs
Nb_3Sn
$PuCoGa_5$
NbN
V_3Si
K_3C_{60}
Li @ 33 GPa
CNT
$PuRhGa_5$
Pb
Nb
diamond
$CeCu_2Si_2$
UBe_{13}
UPt_3
UPd_2Al_3
$CeCoIn_5$
LaOFeP
Hg
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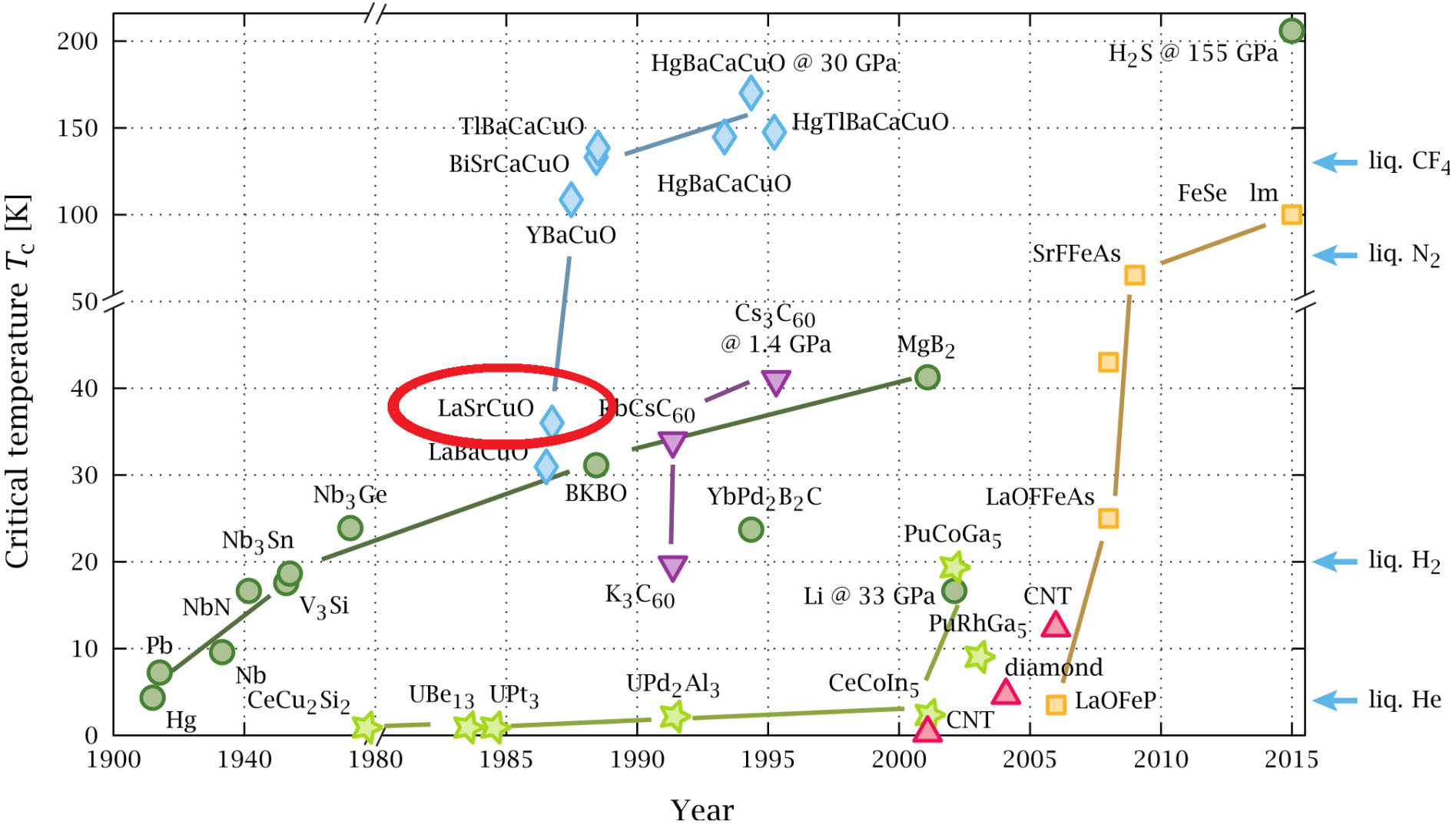
liq. CF_4
liq. N_2
liq. H_2
liq. He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CNT
liq. CF_4
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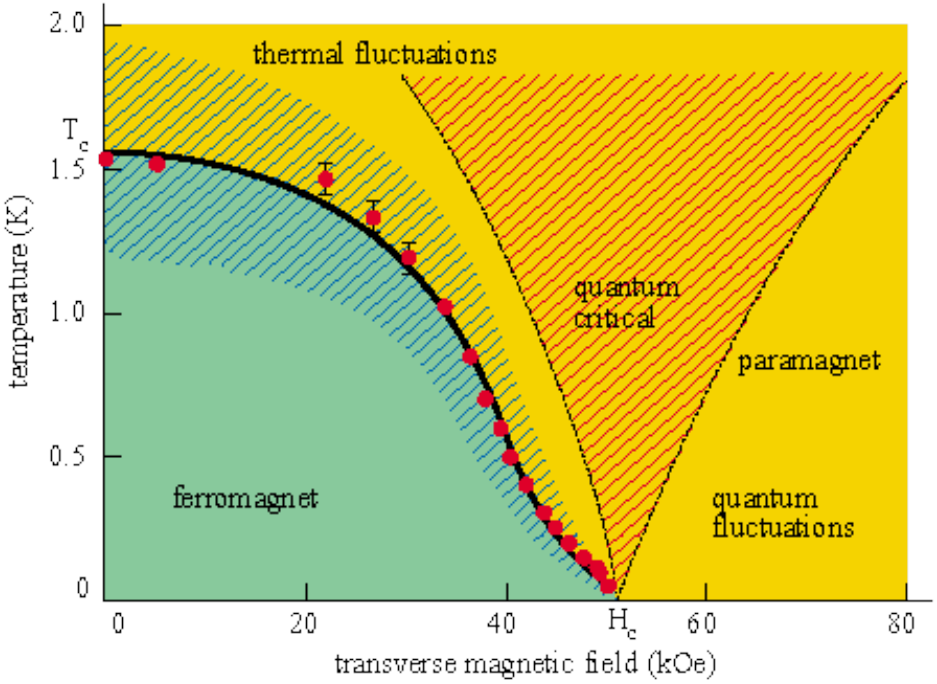
liq. He

Classical vs. Quantum Phase Transitions

Classical Phase Transitions:

- Occur at *finite* temperature T_c.
- Characterized by *order parameter* (e.g. magnetization, Ψ, ..)
- Fluctuations at criticality can be captured by *classical* statistical mechanics (e.g., Landau-Ginsburg theory)

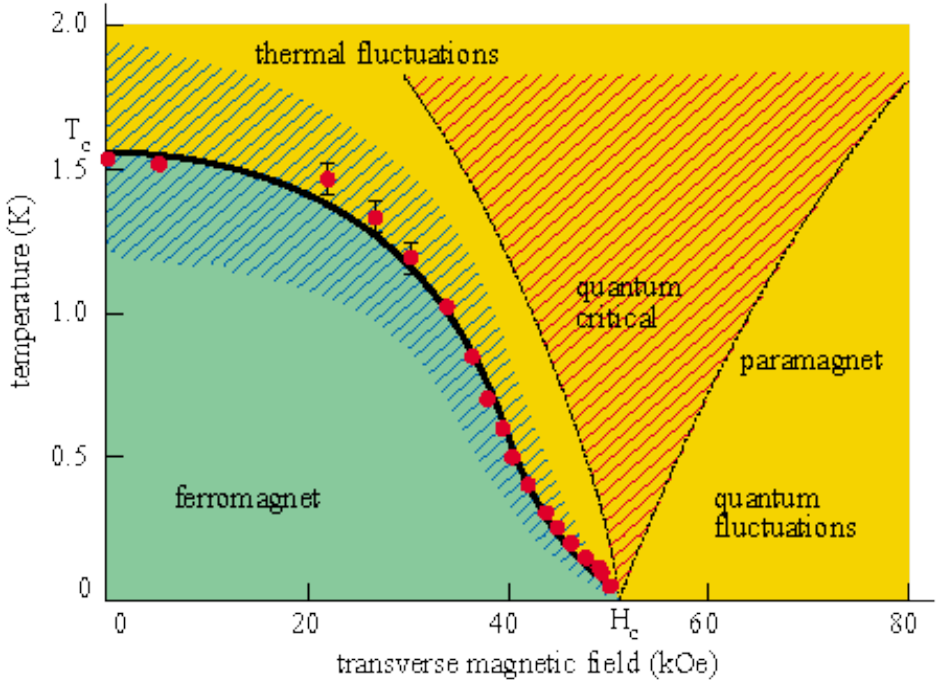


A phase diagram for an Ising ferromagnet, credit: sachdev.physics.harvard.edu/physworld/

Classical vs. Quantum Phase Transitions

Quantum Phase Transitions (QPTs):

- Occur at $T = 0$.
- Brought about by changing tuning parameter in quantum Hamiltonian (film thickness, magnetic field, doping, ...)
- Driven by quantum mechanical, as opposed to thermal, fluctuations



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Quantum Statistical Mechanics and Finite-"Size" Scaling

Statistical Mechanics governed by the partition function:

$$Z(\beta) = \sum_n \langle n| e^{-\beta H} |n\rangle \tag{1}$$

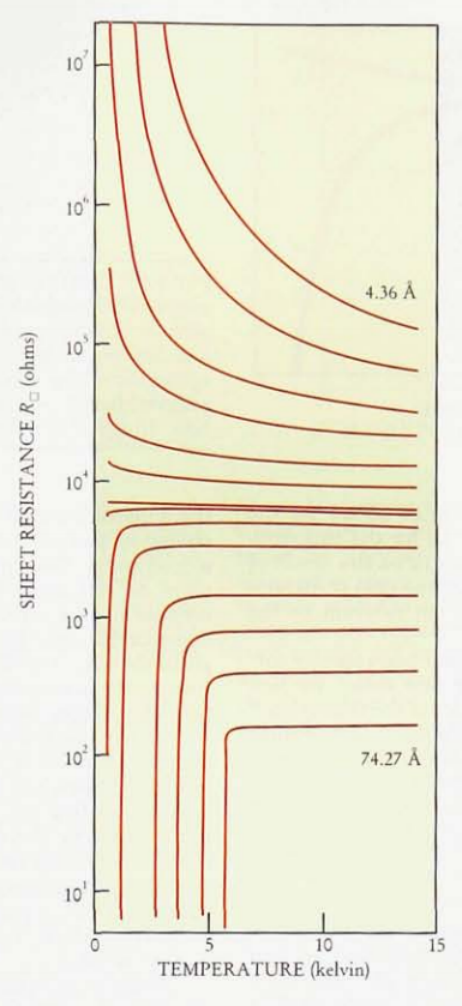
Looks like "imaginary time" evolution if we take $\tau = -i\hbar\beta$:

Allows us to write d-dimensional quantum system as $d+1$ dimensional classical system!

Size of extra dimension depends on β – infinite only at $T = 0$.

For non-zero temperatures (which could destroy the QPT), have to use finite "size" scaling

Superconductor-Insulator Transition as a QPT



Physics Today, **51**(11), 39.

Results of the 90's hinted that some superconducting transitions (driven by disorder or magnetic fields) might be quantum phase transitions.

Finite size scaling makes predictions on resistance as a function of T and the tuning parameter, x:

$$R_\square(x, T) = R_c F(|x - x_c| T^{-1/z\nu}) \qquad (2)$$

Particular models of the QPT make predictions of R_c, z, and ν.

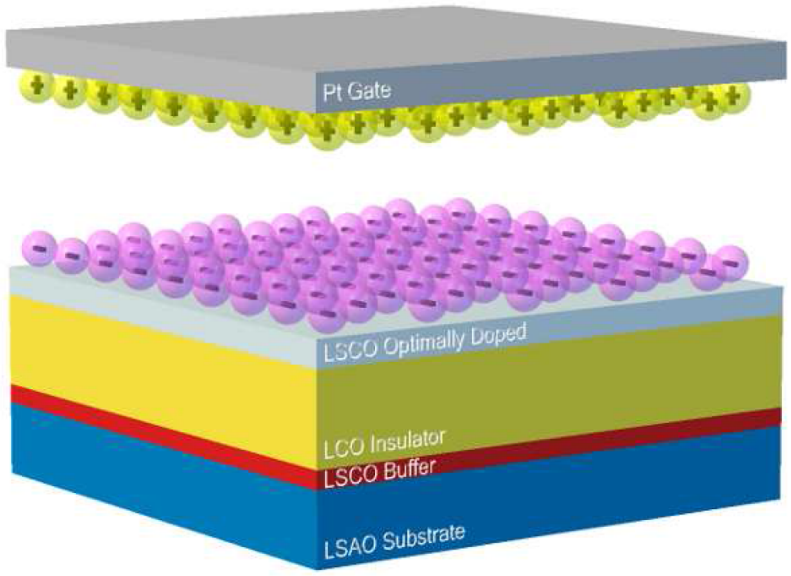
"Dirty Boson" picture predicts $R_c = h/(2e)^2 = 6.45$ kΩ– the pair quantum resistance, R_Q.

Fabrication – The Electric Field Effect

Created ultrathin films of $La_{2-x}Sr_xCuO_4$ using atomic-layer molecular beam epitaxy, between 1 and 2 unit cells thick.

Electric field effect: use voltage to accumulate or deplete charge carriers in material, without introducing disorder.

Varied gate voltage V_G to tune carrier density x by ± 0.04, leading to shifts in T_c up to 30 K.



Schematic of the heterostructure used to fabricate films with different carrier densities.

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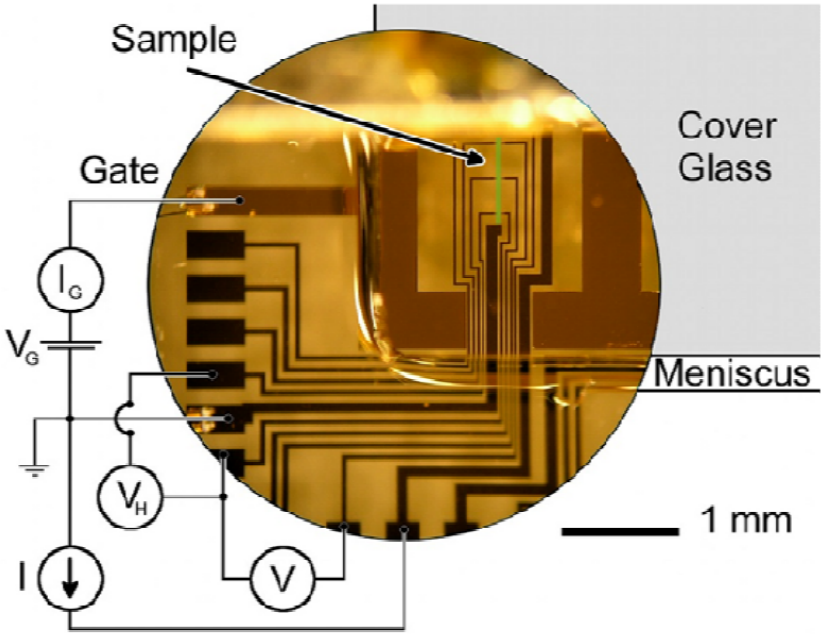
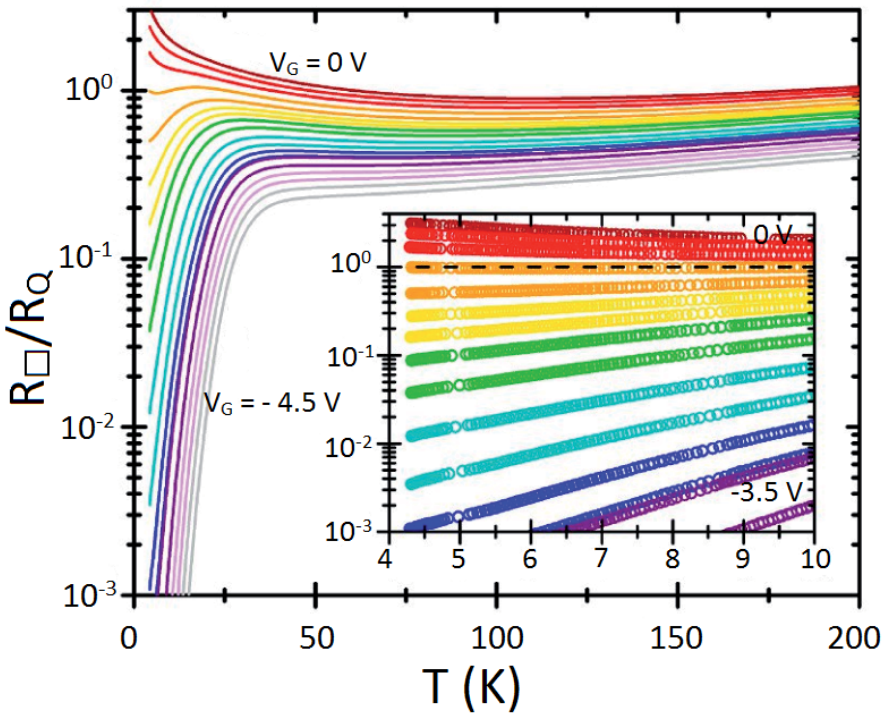
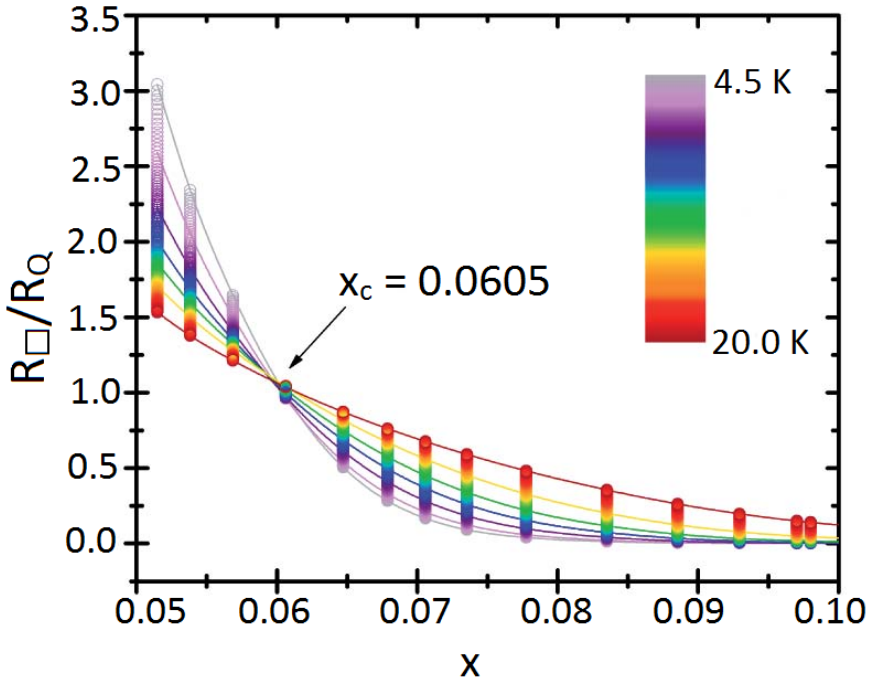


Optical micrograph of a device, with cartoon of the circuits used for measurements.

Scaling Behavior

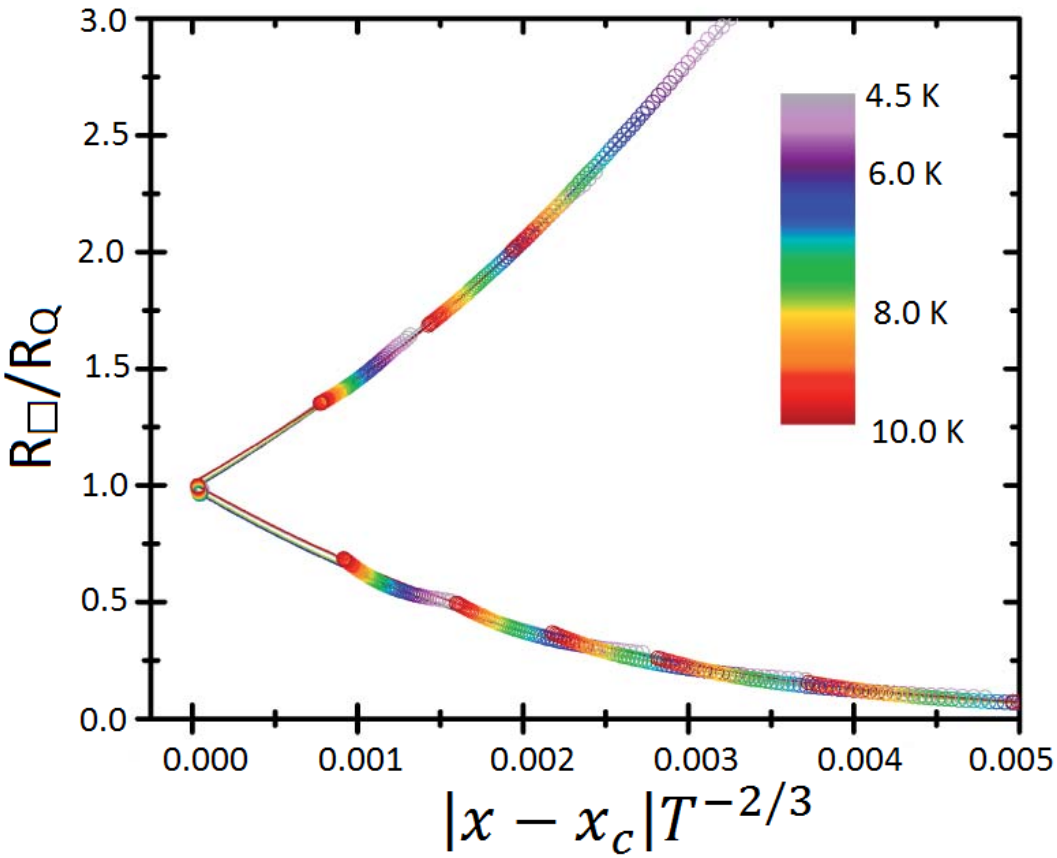
Complementary plots of resistance vs. temperature / carrier density confirm existence of a phase transition.





Collapse

Plotting the resistance against a particular combination of x and T shows a collapse with properly chosen exponents.



The critical resistance is R_Q – as predicted by the dirty boson picture!

Conclusions and Outlook

- Understanding high-T superconductivity is one of the biggest outstanding problems in physics.
- Certain superconductor transitions appear to be quantum phase transitions
- Experiments on Copper Oxide Superconductors prefer bosonic models of the phase transition

Outlook:

- Further work on magnetic-driven transition might confirm bosonic model.
- Potential applicability in higher T_c superconductors

Thanks to Professor Stephens for all his comments and advice!

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